ERRATA: Database Systems Assignment 03

Corrected Logic for Problems 1.4 – 1.10

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1 Problem 1.4: Universal Service Subscribers

The requirement is to find subscribers who have every service listed in the system. This is a classic Relational Division problem.

Relational Algebra

$$\pi_{name,address}(subscribers \bowtie (\gamma_{portid;count(service)\rightarrow c}(service_subscribers) \ltimes \sigma_{c=(\gamma_{count(scode)\rightarrow tc}(services))}(services)))$$

```sql
SELECT s.name, s.address
FROM subscribers s
JOIN service_subscribers ss ON s.portid = ss.portid
GROUP BY s.name, s.address
HAVING COUNT(DISTINCT ss.service) = (SELECT COUNT(*) FROM services
    );
```

2 Problem 1.5: Service Threshold

Find subscribers with at least three services.

Relational Algebra

$$\pi_{name}(subscribers \bowtie \sigma_{count\geq 3}(\gamma_{portid;count(service)\rightarrow count}(service_subscribers)))$$

```sql
SELECT s.name
FROM subscribers s
JOIN service_subscribers ss ON s.portid = ss.portid
GROUP BY s.name
HAVING COUNT(ss.service) >= 3;
```

3 Problem 1.6: Most Popular Service

Identifies the service(s) with the maximum number of subscribers.

Relational Algebra

Let $R = \gamma_{service;count(*)\rightarrow c}(service_subscribers)$
Result: $\pi_{service}(\sigma_{c=max(c)}(R))$

```sql
SELECT service FROM service_subscribers
GROUP BY service
HAVING COUNT(*) = (
    SELECT MAX(cnt) FROM (SELECT COUNT(*) AS cnt FROM
    service_subscribers GROUP BY service)
);
```

4 Problem 1.7: Trunk Routing Preference

Implements longest-prefix matching for area code 416 and office 334.

Relational Algebra

$\pi_{portid,foreign_switch}(\sigma_{(area='416' \wedge office='334') \vee (area='416' \wedge office='000') \vee (area='000' \wedge office='000')}(trunks \bowtie trunk_routes))$

```sql
SELECT t.portid, t.foreign_switch
FROM trunks t JOIN trunk_routes tr ON t.portid = tr.portid
WHERE (tr.area = '416' AND tr.office = '334')
   OR (tr.area = '416' AND tr.office = '000')
   OR (tr.area = '000' AND tr.office = '000')
ORDER BY CASE
    WHEN tr.area = '416' AND tr.office = '334' THEN 1
    WHEN tr.area = '416' AND tr.office = '000' THEN 2
    ELSE 3 END ASC;
```

5 Problem 1.8: Line Availability

Checks the IDLE state for a specific directory number.

Relational Algebra

$\pi_{portid,state}(\sigma_{area='613' \wedge office='712' \wedge number='0024'}(lines \bowtie channels))$

```sql
SELECT l.portid, '(' || l.area || ') ' || l.office || '-' || l.
    number AS directory_number, c.state
FROM lines l JOIN channels c ON l.portid = c.portid
WHERE l.area = '613' AND l.office = '712' AND l.number = '0024';
```

6 Problem 1.9: Routing with Availability

Same as 1.7 but ensures at least one channel is 'IDLE'.

```sql
SELECT DISTINCT t.portid, t.foreign_switch
FROM trunks t
JOIN trunk_routes tr ON t.portid = tr.portid
JOIN channels c ON t.portid = c.portid
WHERE (routing_criteria) AND c.state = 'IDLE'
ORDER BY preference_logic;
```

7 Problem 1.10: Jason Allison Subset

Finds subscribers who have at least all the services of Jason Allison (portid 2).

Relational Algebra

$Subscribers \div (\pi_{service}(\sigma_{portid=2}(service_subscribers)))$

```sql
SELECT s.name FROM subscribers s
WHERE NOT EXISTS (
    SELECT ss1.service FROM service_subscribers ss1 WHERE ss1.
    portid = 2
    EXCEPT
    SELECT ss2.service FROM service_subscribers ss2 WHERE ss2.
    portid = s.portid
);
```